

1) Stack and Queue

Stack:

Push(54) $\rightarrow$ (54)Push(52) $\rightarrow$ [54, 52]Pop() $\rightarrow$ remove 52push(55) $\rightarrow$ [54, 55]push(62) $\rightarrow$ [54, 55, 62]S.pop() $\rightarrow$ 62

Queue:

• enqueue(21) $\rightarrow$ [21]• enqueue(24) $\rightarrow$ [21, 24]• dequeue() $\rightarrow$ remove 21• Enqueue(28) $\rightarrow$ [24, 28]• Enqueue(32) $\rightarrow$ [24, 28, 32] $q = \text{dequeue}() \rightarrow 24$ $q = 24$ $S + q = 62 + 24 = 86$ 2) Exp, $105 + 606 / * 8 -$ • $10 + 5 = 15$ • $60 / 6 = 10$ • $15 \times 10 = 150$ • $150 - 8 = 142$

Ans: 142

3) Postfix $8\ 2\ 3^{\wedge} / 2\ 3^* + 5\ 1^* -$

• $2\ 3^{\wedge} = 8$

• $8/8 = 1$

• $2\ 3^* = 6$

• ~~1~~ $(6, 1)$

4) Assume op

Infix : $a + b * c - d^{\wedge} e^{\wedge} f$

postfix = $abc * + def^{\wedge} \wedge -$

5) Push(10) Push(20) pop(), push(10), Push(20),
pop(), pop(), pop(), push(20), pop()

$\rightarrow 10, 20, 20 \Rightarrow 20, 20, 10, 10, 20$

$\rightarrow 20$

Stacked Value $\Rightarrow \{20, 20, 10, 10, 20\}$

4)

Character	Stack	op
A	Empty	A
+	+	A
B	+	AB
*	+	AB
C	+	ABC
-	-	ABC*
d	-	ABCD

n	- n	ABC * + d
e	- n	ABC * + de
n	- nn	ABC * + de
f	- nn	ABC * + def
-	- n	ABC * + def nn
empty	empty	ABC * + def nn -

29/12